MILITARY SPECIFICATION

SYRINGE, FOUNTAIN, BARIUM ENEMA. DISPOSABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of completely assembled, disposable barium enema bag.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566 Boxes, Folding, Paperboard.
PPP-B-636 Box, Fiberboard.
PPP-B-676 Box, Setup.

Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

FSC 6530

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Pharmacopoeia of the United States

(Applications for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Plastic. The plastic sheeting shall be polyvinyl calendered sheet, dull finish, clear stock. The plastic tubing shall be of transparent vinyl. The sheeting and tubing shall contain not more than 5 parts per million of lead when tested as specified in 4.4.1. In addition, the plastic shall pass the toxicity test specified in 4.4.2.

3.2 Design and construction. The syringe shall consist of a leakproof plastic bag, plastic tubing with integral rectal tip and a shutoff clamp. Construction shall be in accordance with either Figure 1 or Figure 2.

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3.2.1 Plastic bag. The plastic shall be heat sealed to form a leakproof plastic bag and shall include a filling spout and an outlet. The bag shall have a minimum capacity of 2500 milliliters (ml.). The plastic material shall be not less than 0.006 inch nor more than 0.012 inch in thickness. One side of the bag shall have graduation markings at 500 ml. intervals at not less than 5 points, 500, 1000, 1500, 2000 and 2500 ml. Markings shall be accurate to ± 25 ml. The opening of the filling spout shall allow filling from a faucet. The opening at the top shall have a suitable leakproof, mechanical or self sealing type closure. The filling spout shall be flexible and pliable but rigid enough to maintain its shape when being grasped to be filled. The top outer edge of the bag (outside of the sealed edge and immediately adjacent to the filling opening) shall be provided with a reinforced opening to permit suspension of the filled bag from an intravenous stand or other support.

3.2.2 Plastic tubing. The tubing shall be sterilizable, have a minimum inside diameter of 0.25 inch, a minimum wall thickness of 0.035 inch and shall not be less than 72 inches long. The tubing shall be flexible and pliable but rigid enough to prevent collapse during normal usage. The tubing shall be permanently attached at one end to the bottom opening of the bag and there shall be no constriction of the opening as a result of the attachment of the tubing. The other end of the tubing shall have as an integral part, an adult rectal tip made from polyethylene. The rectal tip shall be as shown on Figure 1, and a standard bore. The finish of the rectal tip shall be smooth and polished.

3.2.3 Clamp. The syringe shall be provided with a positive acting on-off clamp, attached to the tubing, which by compression of the tubing shall be capable of accomplishing complete shut off. The bags shall show no leakage nor hanger rupture when tested as specified in 4.4.3 and 4.4.4. The on-off clamp shall be of chrome plated steel.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Clamp test	3.2.3	4.4.3	S2	*
Drop test	3.2.3	4.4.4	S2	*

*The acceptance number shall be zero, and not less than three samples shall be taken regardless of lot size.

4.2.3 Component materials. Sufficient sample of plastic material shall be selected from each lot used in the manufacture of the syringe to perform the tests specified in 4.4.1 and 4.4.2. A lot of plastic tubing or sheeting shall be defined as one continuous production run using the same batch of ingredients.

4.3 Examination. The syringes shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, to determine compliance with specification requirements.

4.4.1 Lead test. Ten grams of sample shall be ignited to char, then further ignited in a muffle furnace at 700° centigrade (C.) to a white ash. The ash shall be boiled in 15 ml. of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion equivalent to one gram of sample shall be placed in a separator. The dithizone method for lead, as described in the current edition of the Pharmacopeia of the United States using 5 parts per million of lead solution as a control, shall be applied.

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4.4.2 Toxicity. Two 2-inch squares of plastic sheeting and two 2-inch lengths of plastic tubing shall be placed in separate containers containing 100 ml. of Sodium Chloride Injection, U.S.P. The solution shall be autoclaved for 35 minutes at 235 degrees Fahrenheit. 0.5 ml. of the solution so prepared shall be cooled and injected intravenously into each of five healthy mice weighing between 17 and 23 grams. At the end of 4, 24, and 48 hours, the animals shall show no discernable symptoms of toxicity. If any of the animals show gross signs of toxicity or die, the test shall be repeated with five unused mice weighing 20 ± 1 gram each. All of the animals shall survive for 48 hours.

4.4.3 Clamp test. Sample syringes with clamps closed shall be filled to the 2500 ml. mark with water at 120°F. and suspended by the hangers for 60 minutes.

4.4.4 Drop test. Sample syringes with clamps and spouts closed and filled with 2500 ml. of water shall be dropped from a height of two feet.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Twenty-four (24) syringes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	NOT APPLICABLE	2 units

5.1.2.1 Packaging variation permitted. If the required number of units to be shipped is less than the number of units specified to be over-packed in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Individual package. One complete syringe, fountain, barium enema, disposal bag shall be packaged in a polyethylene bag of sufficient thickness or shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. The syringe tubing shall be formed into a coil and shall be kink free. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be constructed in accordance with PPP-B-636, type CF, class domestic, and method I closure shall be utilized.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual package. Each individual package shall be marked to identify contents.

5.3.2 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Rectal administration of barium in conjunction with roentgenography of the intestinal tract.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packing (see 5.1.3 and 5.1.4).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
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Preparing activity:
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category N

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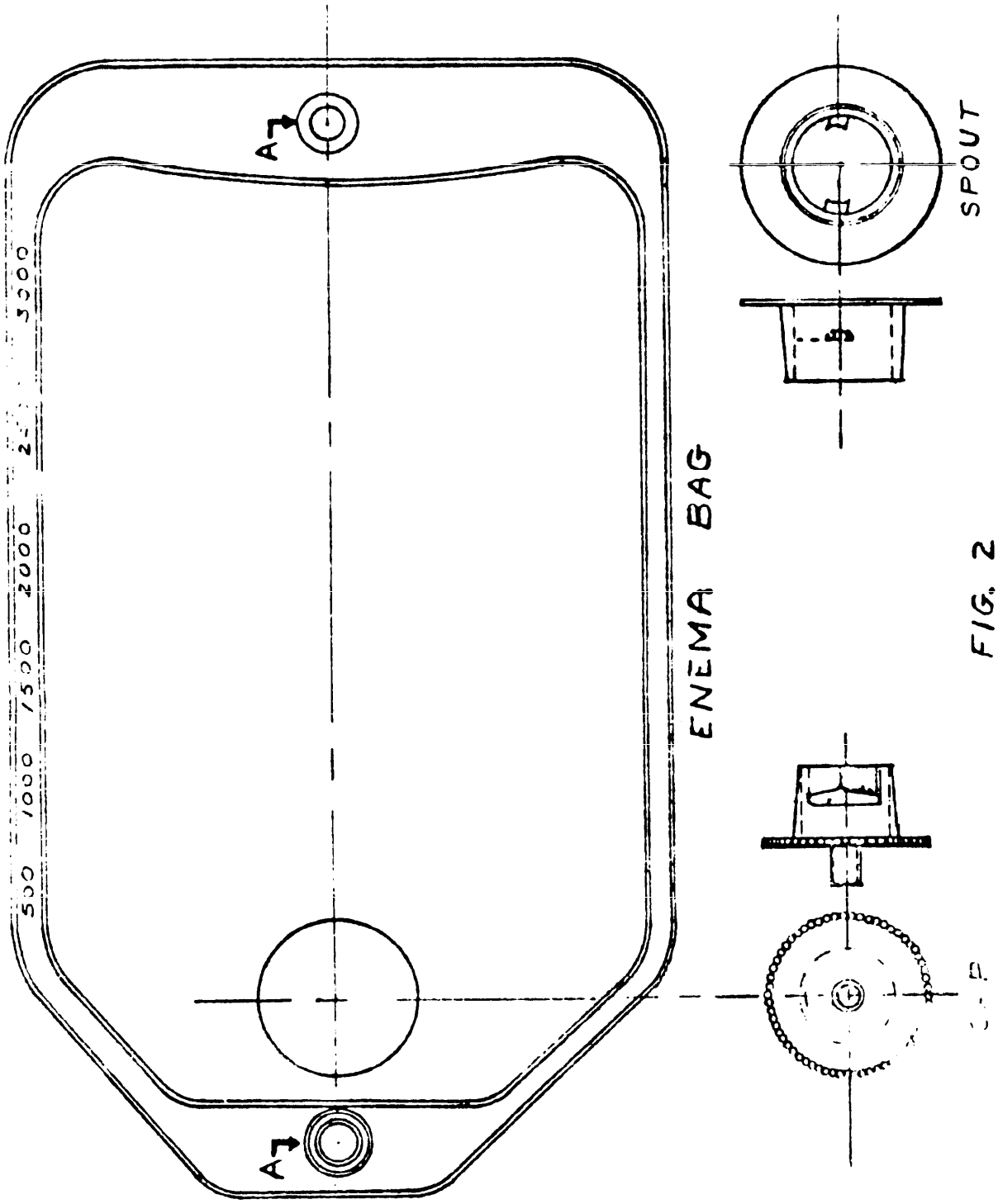


FIG. 2

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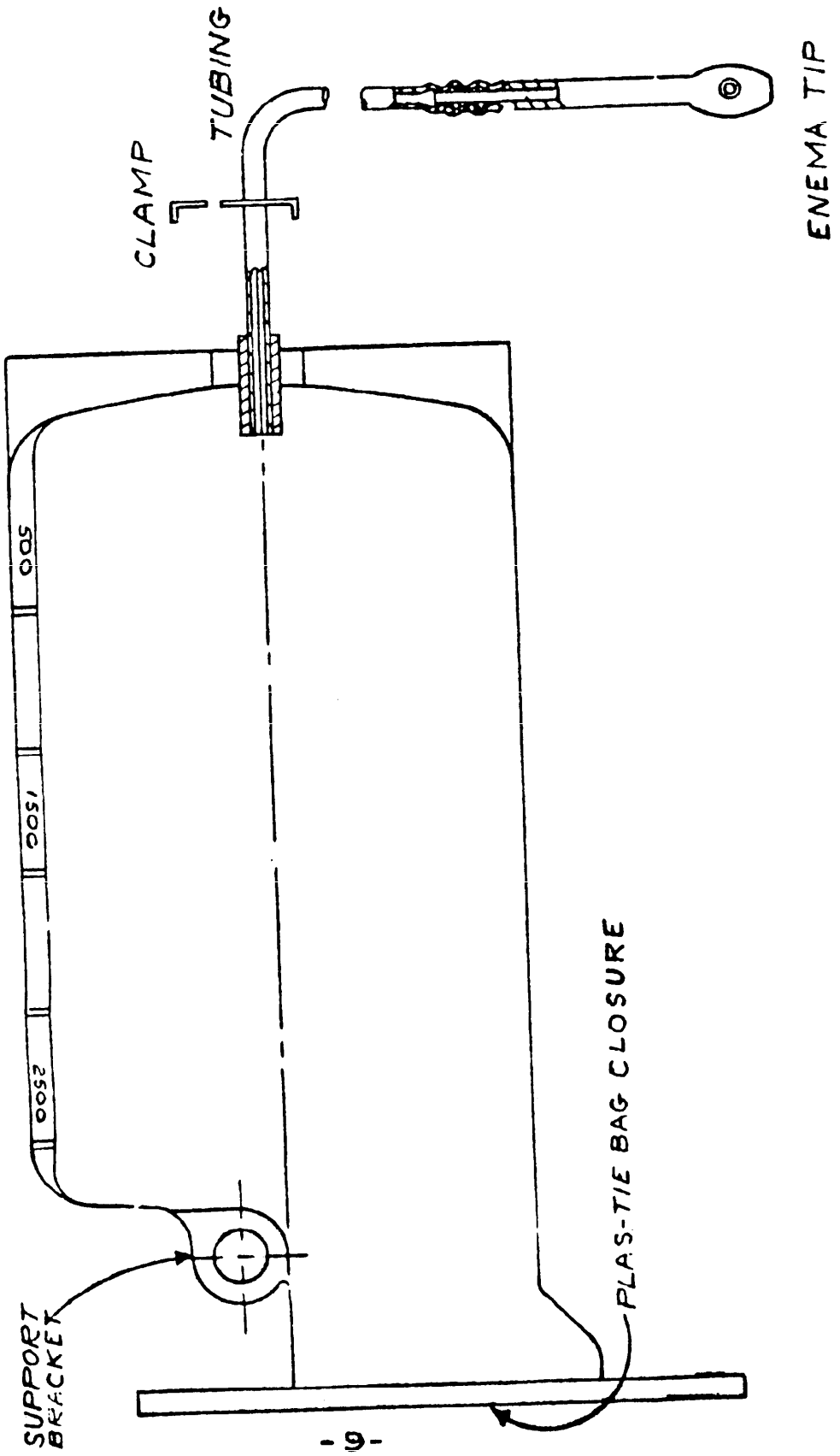


FIG. 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. [illegible]

INSTRUCTIONS

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products are procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity as indicated on reverse sheet.

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN THIS AREA? YES

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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PLATE NO. 15419 (BACK)

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